Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Biocide-free sterilization of process fluids through a combination of shear force and UV-based methods*			
Korea's 12 National Strategic Technology Fields	**Advanced Bio, Advanced Robotics & Manufacturing, Others**			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Energy/ Environment Mechanical Systems	*Water Treatment Equipment, Waste Treatment Equipment*	*Surface Treatment*	*Other Surface Treatment Technologies (Pretreatment)*
Concept of Proposed Technology	*Microbial contamination of water-carrying systems presents a challenge to smooth operation and user safety. A standard method for the elimination of microorganisms is the addition of biocides. Since these not only pollute the environment but can also cause microorganisms to develop resistance, it is advisable to reduce the use of these materials. The proposed project aims to develop a sterilization process that uses a shear force-based method (e.g., rotor-stator-based homogenizer or flow-dynamic water treatment) in combination with UV light to significantly reduce the bacterial count. The aim is to eliminate bacteria, both thin- and thick-walled, as well as fungi and their spores. Continuous treatment in the bypass should prevent the formation of stubborn biofilms and avoid the excessive addition of biocides or, in the best case, make them superfluous. The release of materials that are harmful to health and the environment as well as damage to the process of fluids must be avoided. The process should be suitable for a wide range of applications.*			

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Rationale for Joint R&D Project	*Microbial contamination is not only a significant problem in drinking water, but can also have a considerable impact on processing fluids:* *So-called process water is used in many industrial sectors, such as the paper industry, the food and drink industry (e.g. meat processing, roasting and smokehouses), the pharmaceutical industry, coating plants (e.g. paint shops) and electronics manufacturing. It is used for cleaning products or equipment, producing beverages and food, cooling systems and transporting products within the process. Its contamination with microorganisms not only poses a health risk to employees in the relevant industry but can also have a negative impact on manufacturing processes and, especially in the hygiene sector, on product quality In the food and beverage industry, for example, contamination of the product by germ-contaminated process water can lead to food poisoning caused by bacteria such as salmonella, pseudomonas and listeria. The number of reported salmonella infections in Germany alone was 13174 in 2023. Furthermore, microbial contamination of process water is difficult to detect because many bacteria do not grow on artificial media, such as those used in contamination testing via dip slides, and therefore often remain undetected. For reasons of sustainability, among other things, process water is often treated after use and then returned to the process, creating a cycle in which microorganisms can multiply more rapidly.* *As a preventive measure, biocides are therefore often used in excessive doses, which is associated with considerable health risks and environmental pollution as well as increased material costs. The need for demand-based dosing of effective biocides is therefore becoming increasingly urgent, particularly due to the increasing incidence of resistance in microorganisms. However, this is complicated by the tightening of labeling limits for sensitizing active substances in the Biocidal Products Regulation, which imposes an obligation on end users to monitor certain active substances. It is to be expected that this will reduce the range of biocidal active substances used by end users in the future. This will also restrict the use of alternative biocidal active substances.* *Therefore, the development of resistance in microorganisms can have more serious consequences if no suitable alternative biocides are available. In summary, it can be said that the use of biocides is not only costly but also unsustainable and subject to strict regulations. Physical disinfection methods are therefore becoming increasingly important. These methods include the application of shear forces, plasma treatments, and UV treatments. While shear forces mainly affect Gram-negative bacteria, the other two methods are known for their broad-spectrum effect. However, there is significant potential for improvement in terms of intensity, power consumption, and treatment duration.* *A promising solution for optimizing the sterilization effect and sustainability is to combine the various physical sterilization techniques. A combination of methods offers several advantages: Combining the methods reduces the number of UV lamps required, which leads to efficient use of resources. Power consumption is reduced because the different processes complement each other and thus work more efficiently. Combining the processes increases the effectiveness of disinfection, as different methods work synergistically and enable more comprehensive germ control.* *Implementing this combination technology in the form of a bypass solution requires less space, which is particularly advantageous in confined application areas and offers the possibility of integrating it*

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	into existing systems without major conversion work. The advantages in terms of resource conservation, power savings, effectiveness, and speed make this method an attractive option for a wide range of industries.
Key Words	1. *Biocide-free sterilization* 2. *Microbial contamination* 3. *shear force-based method* 4. *UV treatment*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	*The aim of the proposed cooperation project is to develop a sterilization device for the biocide-free or low-biocide sterilization of aqueous media, such as process fluids, by combining shear force and UV-based methods. The aim is to achieve a significant reduction in microorganisms, eliminating both Gram-negative and Gram-positive microorganisms as well as fungi and spores. Both sterilization methods are designed to attack unwanted biological contamination in aqueous media in different ways. In the first stage, Gram-negative microorganisms are broken down and killed by shear forces. This involves the use of methods such as a rotor-stator homogenizer or flow-dynamic water treatment, both of which can break down cells on the basis of shear forces and pressure changes. This effect has already been observed in previous projects, but it almost exclusively eliminates thin-walled Gram-negative bacteria in terms of the cell wall. With an integrated second stage consisting of ultraviolet light irradiation, which damages the DNA of gram-positive cells, fungi, and fungal spores, thereby preventing cell proliferation. Figure 1 shows a schematic representation of the structure and operating principle of this innovative development in bypass technology, which can be flexibly and cost-effectively integrated into existing process technology.*			

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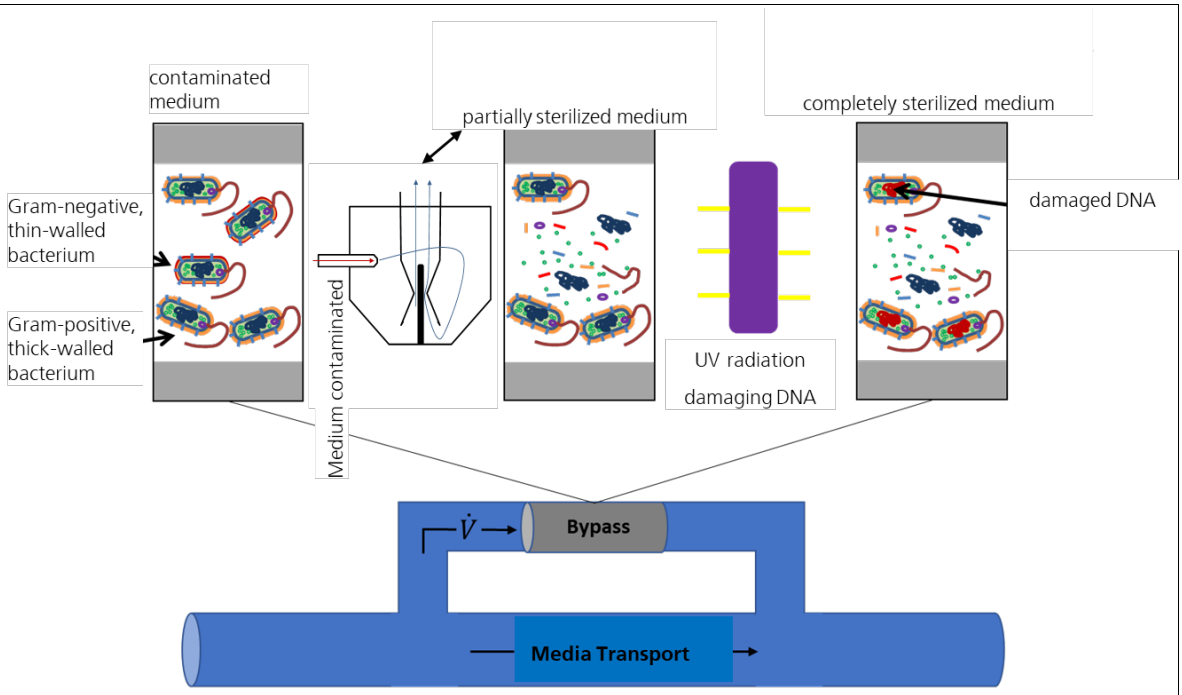

Figure 1: Schematic representation of the structure and operating principle of the development in the bypass guide: The existing Gram-negative microorganisms are broken down and killed by shear forces. A further process step using UV light alters the DNA of the Gram-positive microorganisms that were not killed by the sheer force, thus preventing the cells from multiplying.

Amination is one of the most feared disruptions, as it impairs coating quality, requires time-consuming cleaning processes, and can lead to total failure of the paint shop. In cathodic electrodipcoating plants (KTL plants), for example, microbial disinfection is essential. Due to the size of the dip coating tanks, refilling the plant with KTL coating material in conjunction with production downtime costs millions. Direct sterilization of the KT paint is not considered in this funding project, but rather sterilization of the subsequent rinses, which use a cascade system to feed the rinse water into the KTL tank, thereby introducing microbiological contamination from the rinse tanks into the KTL tank.
There is also the possibility of integrating the sterilization system to be developed into various stages of the pretreatment process. Numerous other possibilities for sterilization in a wide variety of areas are possible. The Korean SME is welcome to suggest its own interesting markets here.

Detailed description of the two-stage sterilization process as part of the research project:
First sterilization stage (sterilization using physical forces):
One applicable technology is the hydro-physical water and liquid treatment system from IFEC HP-Tech GmbH, which is based on fluid dynamic water treatment. The core of the system is a reaction tank with a reaction chamber whose internal structure resembles a cyclone (Figure 1, left). A pressure pump forces water into the reaction chamber at high pressure, where high centrifugal and centripetal forces are achieved. In addition, high shear and friction forces, negative pressure, and cavitation are generated. These physical forces tear apart thin-walled, gram-negative bacteria and destroy them. The effectiveness can be proven on the basis of previous projects, although a reduction in the mostly thick-walled, Gram-positive microorganisms, fungi, and their spores is not detectable.

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Second sterilization stage (UV light):

The use of UV light is being considered for the second stage of the new sterilization technology. There are two common approaches to sterilizing process water.

On the one hand, direct sterilization by UV light is used. In this method, the UV light is directed directly at the medium to be sterilized. As a rule, UV-C light with a wavelength of 200–280 nm is used, which has proven to be particularly effective in killing microorganisms. The UV-C light penetrates the cells of bacteria, viruses, fungi, and their spores, causing damage to their DNA. This results in the microorganisms being unable to reproduce, which leads to their death. The sterilizing effect is directly proportional to the time the microbes are exposed to the UV light. Longer exposure results in a higher kill rate. This method is very effective in clear water, but penetration can be impaired by turbidity, suspended solids, or chemical contaminants. For the use of direct UV light disinfection, it must be investigated in which geometries (flat or round media flow) effective disinfection is guaranteed. In addition, the exposure time and the resulting kill rate must be examined in more detail.

Indirect UV disinfection using ozone can be considered the opposite of direct UV light disinfection. The conversion of oxygen molecules (O_2) into ozone (O_3) is achieved by UV light in an aqueous medium. The photolysis of oxygen leads to the formation of atomic oxygen, which reacts with oxygen molecules to form ozone. The bond between the oxygen atoms is broken by UV light; this reaction is shown in reaction equations 1 and 2 below. The resulting ozone then decomposes back into molecular and atomic oxygen, as shown in reaction equation 3. In combination with water, the ozone then decomposes into oxygen (O_2) and hydroxyl radicals (OH), which act as strong oxidizing agents and can also kill microorganisms (see reaction equation below).

1. $O_2 \xrightarrow{h\upsilon} O\cdot + \cdot O$ $\quad h\upsilon < 190$ nm
2. $O_2 + O\cdot \xrightarrow{h\upsilon} O_3$ $\quad h\upsilon < 190$ nm
3. $O_3 \xrightarrow{h\upsilon} O_2 + \cdot O$ $\quad h\upsilon < 190$ nm
4. $O_3 + H_2O \xrightarrow{h\upsilon} O_2 + 2OH\cdot$ $\quad h\upsilon < 190$ nm

Ozone is a powerful oxidizing agent and can attack and damage bacterial cell walls through oxidation, ultimately leading to cell death. In addition, ozone can also penetrate DNA and oxidize it. This prevents reproduction and leads to cell death. Compared to direct UV light, it has a broader antimicrobial effect. The decisive advantage of this method lies in its effectiveness in killing even microorganisms that are difficult to access. It can even penetrate biofilms and other protective layers that surround bacterial agglomerates, for example. The project will specifically investigate how external oxygen input in different concentrations affects ozone formation and the resulting sterilization rate. A corresponding schematic diagram is provided in Figure 2.

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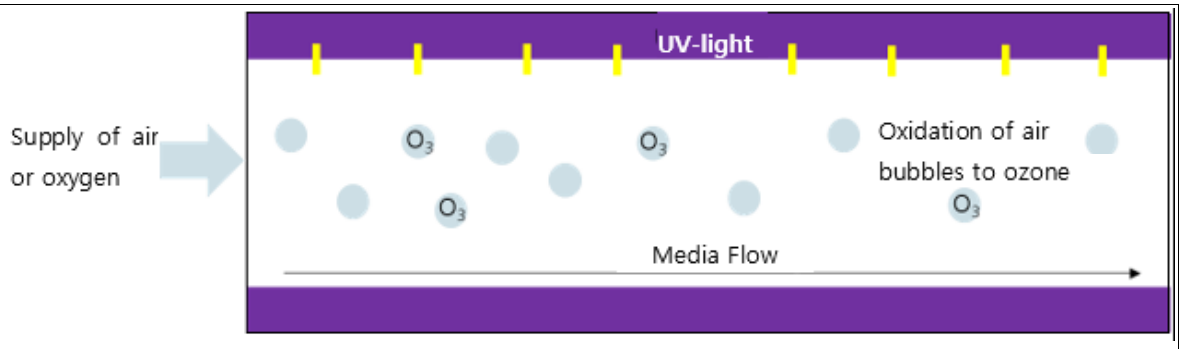

Figure 2: Schematic drawing showing the enhancement of sterilization using ozone through the introduction of external air and/or oxygen into the contaminated medium.

R&D Target Specifications

By combining the three approaches of "shredding," "irradiation with UV light," and "ozone treatment," faster and more effective disinfection can be achieved upon successful project completion than with conventional methods, which also represents a more environmentally friendly alternative, as biocides are reduced or possibly even become obsolete, and the application is more energy-efficient and resource-saving due to optimization in the UV lamp application. The resulting prototype is designed in such a way that the intended bypass solution not only saves space but also requires less maintenance. If the disinfection of industrial water is successful, efforts could be made to transfer it to the drinking water sector, where the daily per capita demand in Germany already exceeds 120 l/day.

R&D Plan

Work plan		1. Year				2. Year			
1	Evaluation of shear force-mediated sterilization methods with the goal of >90% sterilization of g+ MOs	■	■	■					
2	Evaluation of ozone-enhanced sterilization with the aim of achieving an O2-mediated ozone concentration of cO3 = 3 g/m3		■	■					
3	Evaluierung der UV-verstärkten Entkeimung mit dem Ziel einer mind. 40 % Entkeimung g- MOs			■	■				
4	Evaluation of UV-enhanced disinfection with the goal of at least 40% disinfection of g-MOs					■	■		
5	Strategy development for upscaling sterilization technology in the form of a bypass with the goal of sterilizing >90% of all MOs at a throughput of 20,000 m3/h							■	■
6	Interpretation, evaluation, and documentation	■	■	■	■	■	■	■	■

Work packages broken down into sub-work packages

1 Evaluation of various shear force-based sterilization methods including initial sterilization testing on a pilot plant scale

1.1 Analysis and pathogen identification of the microbial load of selected process fluids using microbial methods

1.2 Development of a method for producing a model contaminated process medium with a defined bacterial count of relevant microorganisms according to pathogen identification

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	1.3 Investigation of the shear force-mediated sterilization effect and temperature development using a high-pressure homogenizer on g+ bacteria, g- bacteria, fungi, and spores *1.4 Investigation of the shear force-mediated sterilization effect and temperature development using flow-dynamic water treatment on g+ bacteria, g- bacteria, fungi, and spores* *1.5 Investigation of the shear force-mediated sterilization effect and temperature development using a laboratory dissolver and high-pressure piston system on g+ bacteria, g- bacteria, fungi, and spores* *2 Evaluation of ozone-enhanced sterilization by introducing air with varying oxygen content into a liquid medium* *2.1 Evaluation of the influence of the size of the phase boundary on ozone formation by varying the size of the air bubbles introduced dB* *2.2 Evaluation of ozone-enhanced sterilization through air injection with an oxygen content of cO2 = 20% (corresponding to ambient air)* *2.3 Evaluation of ozone-enhanced sterilization through air injection with an oxygen content of cO2 = 60%* *2.4 Evaluation of ozone-enhanced sterilization through air injection with an oxygen content of cO2 = 90%* *2.5 Introduction of pre-generated ozone from plasma ozone generation* *3 Evaluation of UV-enhanced sterilization and ozone formation by varying the wavelength and geometry of the media flow* *3.1 Variation of the wavelength in the UV-C range using various UV lamps to determine the strongest antimicrobial effect and the optimal wavelength for simultaneous ozone generation, selection of alternating lamps with both wavelengths* *3.2 Investigation of the influence of the dimensions of a flat media guide on the degree of UV disinfection, considering the effective penetration depth of the UV-C rays; influence of the dimensions of a round media guide on the degree of UV disinfection, taking into account the effective penetration depth of the UV-C rays and the temperature development.* *3.3 Testing the sterilization effect by plating out on agar culture media. Here, the number of germs in colony-forming units (CFU)/ml is determined. The sterilization effect is calculated as the quotient of the total germ count after and before treatment.* *4 Development and construction of a modular and therefore flexible functional model for disinfection by combining a selected physical process with UV light and ozone formation* *4.1 Evaluation of the methods to be considered and design of the laboratory demonstrator based on the results from AP 1, AP 2, and AP 3* *4.2 Development and construction of a functional model on a pilot plant scale and investigation of temperature development* *4.3 Microbial investigation of the mechanism of action of combined water disinfection with regard to the different microorganisms during single and multiple passes of the contaminated process medium (gram-negative and gram-positive bacteria as well as fungi and spores)* *5 Strategy for upscaling the further developed sterilization technology in the form of a bypass for*

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	use in industrial plants 5.1 Evaluation of the dimensions and design of the upscaled functional model for the bypass in terms of location, design and size, energy requirements, flow rate, operating time, and temperature based on the results from AP4 5.2 Development and construction of an upscaled bypass prototype 5.3 Conducting sterilization tests and determining the treatment duration with the upscaled bypass prototype in a complete system with a volume V = 20-40,000 m3/h, measured in the bypass. 6 Interpretation, evaluation, and documentation of test results and report generation 6.1 The results obtained are evaluated quantitatively and qualitatively to generate statistical key figures and visual representations; the findings are recorded in the form of a report. Milestones 1. Milestone: Identification of the pathogen environment of selected process fluids and simulation strategy 2. Milestone: Identification of the most effective antimicrobial shear force-mediated sterilization process 3. Milestone: Identification of the optimal air bubble size dB for ozone formation in water through UV radiation 4. Milestone: Oxygen content cO2 with formed ozone quantity cO3 = 5–10 g/m3 was identified and a functional version of the oxygen injection is available 5. Milestone: Functional version of UV water treatment with simultaneous ozone formation identified 6. Milestone: Functional version of the prototype for the combined process available 7. Milestone: Upscaling strategy for the construction of a bypass demonstrator presented 8. Milestone: Final report completed

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The innovative core of Fraunhofer IPA consists of developing a prototype of the bypass sterilization system on a pilot plant scale. The necessary expertise is available thanks to the existing expertise in mechanical engineering and a high level of knowledge of coating technology in the context of surface processes, technology, and materials. Furthermore, Fraunhofer IPA has extensive knowledge of sterilization methods based on oxidative processes, such as dielectric barrier discharge (DBD) or UV-based sterilization methods. In addition, the institute already has equipment for biocide-free sterilization, such as flow-dynamic water treatment.

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Expected R&R of Korean SME Partner	*The focus of the Korean SME's work could, for example, be to merge and scale up the prototype at Fraunhofer IPA, based on their groundwork, to a production-like scale. The jointly developed new sterilization technology is to be transferred to a robust demonstrator that can withstand everyday production. The medium to be sterilized must not damage the demonstrator, and it must be tested to ensure that the demonstrator can withstand long-term testing, for example in conjunction with aggressive chemicals such as moderately strong acids and alkalis, etc.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The aim of the R&D project is to dimension and construct a bypass prototype using the developed process and to evaluate it successfully in an industry-related context at a selected customer's site. Once the R&D project has been completed, development of the production plant, manufacturing, and marketing can begin.* *The following technical target criteria apply for use in chemical industrial plants, such as paint shops, smokehouses, and production facilities in the paper industry:* *• Receive the basis for plant construction in production processes* *• Verification of the functionality of the prototype in practical tests* *• Ensuring the transferability of the prototype to the operating facilities* *• Subsequent distribution and manufacture of the plant technology* *In addition, the following economic target criteria apply to the application* *• Proof of economic efficiency* *• Establishment on the market, including achievement of sales target of €750,000/year* *• Recouping R&D costs through the sale and installation of the systems within 2–3 years of completion of the R&D project* *• Generation of profit from the marketing of the plant technology within 3 years*

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